Confidence, α, critical value, SE and margin of error

C8.01-C8.68 CORE ONLY (SKILL IDs)

1 RECOGNISE IT

- Any confidence-interval question before selecting a parameter-specific method.
- Confidence level C is supplied.
- Need critical value, SE, margin or endpoints.

2 DECISION / METHOD

1. α=1-C.
2. For a two-sided CI, place α/2 in each tail.
3. Select Z or t.
4. Compute SE.
5. Margin e*=critical×SE; CI=estimate±e*.

3 FORMULA

$$CI = \text{estimate} \pm \text{critical} \times SE$$
$$\alpha = 1 - C$$
$$z^* = \Phi^{-1}(1-\alpha/2) > 0$$
$$t^* = T.INV.2T(\alpha, df) > 0$$
$$\text{margin} = \text{critical} \times SE$$

4 FORMULA SHEET VS WHAT I KNOW

Gives: parameter-specific SE formulas.
Still know: general CI structure, confidence-to-α conversion, and when critical is Z versus t.

5 EXCEL ROUTE

- Z critical: `=NORM.S.INV(1-alpha/2)`
- t critical: `=T.INV.2T(alpha,df)`
- endpoints: estimate ± critical*SE.

6 TINY WORKED EXAMPLE

For C=.90, α=.10 and each tail=.05. The positive Z critical is NORM.S.INV(.95)=1.6449.

7 TOP EXAM TRAPS

- Using α instead of α/2 for a two-sided CI.
- Confusing confidence level with critical value.
- Using SD where SE is required.

8 SAY ALOUD

Every confidence interval is point estimate plus or minus a critical value times a standard error.

9 EXIT CHECK

For 98% confidence, what cumulative-left area gives the positive Z critical?

ONE-GLANCE VISUAL ROUTE

